

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 March 2023

22MCA105 - SOFTWARE ENGINEERING AND TESTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit - I	Marks	BT*	CO*	PO*
1.	a)	Explain Software Engineering, with a neat diagram, explain briefly the waterfall Model.	10	L2	1	2
	b)	Explain Software Evolution.	10	L2	1	1
2.	a)	Explain briefly the Incremental Software process Model with a neat diagram.	10	L2	1	2
	b)	Explain the Legacy System Components with a neat diagram.	10	L2	1	1
		Unit - II				
3.	a)	Explain the Non-functional requirements of Software Engineering with examples.	10	L2	2	2
	b)	Describe Requirement Elicitation and Analysis Process.	10	L2	2	2
4.	a)	Outline the Spiral view of the Requirements Engineering Process.	10	L2	2	2
	b)	Explain Requirement Validation.	10	L2	2	2
		Unit - III				
5.	a)	With the help of State Diagram explain the working of Microwave Oven.	10	L2	3	2
	b)	Define an Activity Model of Insulin Pump Operation in Data driven Model.	10	L2	3	2
6.	a)	Explain Architecture Design.	10	L2	3	2
	b)	What is Context Model? Explain the context of ATM Model.	10	L2	3	2
		Unit - IV				
7.	a)	Explain the six essentials of Software Testing.	10	L2	4	2
	b)	Explain three critical success factors for implementing verification.	10	L2	4	2
8.	a)	Explain validation activities in detail.	10	L2	4	2
	b)	Explain black box and white box test methods with examples.	10	L2	4	2
		Unit - V				
9.	a)	Explain Low level Testing and High level Testing.	10	L2	5	1
	b)	Explain Structural design elements.	10	L2	5	2
10.	a)	Explain approaches to organizing the test function.	10	L2	5	1
	b)	Write short notes on				
		i) Testing is each person's responsibility				
		ii) Testing is performed by dedicated resource.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
